



DATABASE PROJECT

Course Title: Database Systems

Project Title: Railway Management System

Group No: 4

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Railway Management System

Code:

```
create database PakistanRailwayDB;  
use PakistanRailwayDB;  
drop database PakistanRailwayDB;
```

```
create schema core;  
create schema booking;  
create schema finance;  
create schema operations;  
create schema staff;  
create schema maintenance;  
create schema cargo;  
create schema amenities;
```

```
drop table core.zones  
create table core.zones (  
    zone_id int identity(1,1) primary key,  
    zone_code varchar(10) not null unique,  
    zone_name varchar(100) not null,  
    headquarters varchar(100) not null,  
    total_stations int,  
    total_route_km decimal(8,2),  
    established_year int check (established_year >= 1850 and established_year <=  
2100)  
);
```

```
create table core.stations (  
    station_id int identity(1,1) primary key,  
    station_code varchar(10) not null unique,  
    station_name varchar(150) not null,  
    city varchar(100),  
    province varchar(100),  
    zone_id int references core.zones(zone_id),  
    total_platforms int,  
    latitude decimal(10,7),  
    longitude decimal(10,7),  
    category varchar(5) check (category in ('a1','a','b','c','d','e'))  
);
```

```
create table core.trains (  

```

```

train_id int identity(1,1) primary key,
train_number varchar(10) not null unique,
train_name varchar(100) not null,
train_type varchar(20) check (train_type in
('express','passenger','freight','special','superfast','local','mail')),
total_coaches int,
max_speed_kmph decimal(5,2),
status varchar(20) check (status in ('active','retired','maintenance')),
zone_id int references core.zones(zone_id),
introduced_year int
);

```

```

create table core.routes (
route_id int identity(1,1) primary key,
route_name varchar(100) not null,
origin_station_id int references core.stations(station_id),
destination_station_id int references core.stations(station_id),
total_distance_km decimal(8,2),
total_stations int,
route_type varchar(20) check (route_type in ('direct','via')),
is_active bit default 1
);

```

```

create table core.train_schedule (
schedule_id int identity(1,1) primary key ,
train_id int references core.trains(train_id),
station_id int references core.stations(station_id),
stop_number int,
arrival_time time,
departure_time time,
halt_minutes int,
distance_from_origin decimal(8,2),
platform_number int,
day_number int default 1 check (day_number >= 1 and day_number <= 10)
);

```

```

create table Core.platforms (
Platform_Id int identity(1,1) primary key,
station_id int references core.stations(station_id),
platform_number int,
platform_length_m decimal(6,2),
platform_type varchar(20) ,
has_roof bit default 1,
current_status varchar(20) check (current_status in
('available','occupied','maintenance'))
);

```

```
create table booking.travel_classes (  
class_id int identity(1,1) primary key,  
class_code varchar(5) not null unique,  
class_name varchar(50) not null,  
fare_multiplier decimal(4,2),  
is_ac bit default 0,  
berths_per_coupe int,  
description varchar(300)  
);
```

```
create table booking.passengers (  
passenger_id int identity(1,1) primary key,  
full_name varchar(150) not null,  
date_of_birth date,  
gender varchar(10) check (gender in ('male','female','other')),  
mobile_number varchar(15),  
email varchar(100),  
cnic_number varchar(20),  
city varchar(100),  
created_at datetime default getdate(),  
is_active bit default 1  
);
```

```
create table booking.bookings (  
booking_id int identity(1,1) primary key,  
pnr_number varchar(20) not null unique,  
passenger_id int references booking.passengers(passenger_id),  
train_id int references core.trains(train_id),  
journey_date date,  
booking_date datetime default getdate(),  
origin_station_id int references core.stations(station_id),  
destination_station_id int references core.stations(station_id),  
class_id int references booking.travel_classes(class_id),  
total_fare decimal(10,2),  
booking_status varchar(20) check (booking_status in  
('confirmed','rac','waitlist','cancelled')),  
booking_channel varchar(20) check (booking_channel in  
('online','counter','agent'))  
);
```

```
create table booking.tickets (  
ticket_id int identity(1,1) primary key,  
booking_id int references booking.bookings(booking_id),  
passenger_name varchar(150) not null,  
age int,
```

```

gender varchar(10) check (gender in ('male','female','other')),
coach_number varchar(10),
seat_number varchar(10),
berth_type varchar(10) check (berth_type in
('lb','mb','ub','sl','su','window','seat','na')),
ticket_status varchar(20) check (ticket_status in
('confirmed','rac','waitlist','cancelled')),
concession_type varchar(50) check (concession_type in ('none','senior
citizen','student','disabled','military','press')),
fare_amount decimal(8,2)
);

```

```

create table booking.waitlist (
waitlist_id int identity(1,1) primary key,
booking_id int references booking.bookings(booking_id),
train_id int references core.trains(train_id),
journey_date date,
class_id int references booking.travel_classes(class_id),
waitlist_number int,
waitlist_type varchar(10) check (waitlist_type in ('wl','pqwl','rlwl','gnwl')),
created_at datetime default getdate(),
current_status varchar(20) check (current_status in
('active','confirmed','dropped'))
);

```

```

create table booking.cancellations (
cancellation_id int identity(1,1) primary key,
booking_id int references booking.bookings(booking_id),
cancelled_by int references booking.passengers(passenger_id),
cancellation_date datetime default getdate(),
reason varchar(300),
original_fare decimal(10,2),
cancellation_charge decimal(8,2),
refund_amount decimal(10,2),
refund_status varchar(20) check (refund_status in
('pending','processed','failed'))
);

```

```

create table finance.fare_rules (
fare_rule_id int identity(1,1) primary key,
class_id int references booking.travel_classes(class_id),
train_type varchar(50),
base_fare_per_km decimal(6,4),
reservation_charge decimal(6,2),
superfast_charge decimal(6,2),
tax_percent decimal(4,2),

```

```
effective_from date,  
effective_to date  
);
```

```
create table finance.concessions (  
concession_id int identity (1,1) primary key ,  
concession_name varchar (100) NOT NULL,  
discount_percent decimal(5,2),  
eligible_classes varchar(100),  
min_age int,  
max_age int,  
required_document varchar(100),  
is_active bit default 1  
);
```

```
create table finance.payments (  
payment_id int identity(1,1) primary key,  
booking_id int references BOOKING.BOOKINGS(booking_id),  
payment_date datetime default getdate(),  
amount decimal(10,2),  
payment_method varchar(25)check (payment_method IN  
('EasyPaisa','JazzCash','Bank Transfer','Cash','Debit Card','Credit Card')),  
transaction_id varchar(100),  
payment_status varchar(20)check (payment_status IN  
('Success','Failed','Pending','Refunded')),  
gateway_name varchar(50),  
gateway_response varchar(200)  
);
```

```
create table finance.refunds (  
refund_id int identity(1,1) primary key,  
payment_id int references finance.payments(payment_id),  
cancellation_id int references booking.cancellations(cancellation_id),  
refund_amount decimal(10,2),  
refund_method varchar(30)check (refund_method IN ('Original Source','Bank  
Transfer','EasyPaisa','JazzCash','Cash Refund')),  
initiated_at datetime,  
processed_at datetime,  
refund_status varchar(20)check (refund_status IN  
('Pending','Processed','Failed')),  
bank_reference varchar(100)  
);
```

```
create table operations.locomotives (  
loco_id int identity(1,1) primary key,  
loco_number varchar(20) NOT NULL,
```

```

    loco_class varchar(20) NOT NULL,
    traction_type varchar(10) check (traction_type IN ('Electric', 'Diesel')) NOT
NULL,
    max_speed_kmph int,
    horsepower int,
    home_shed_id int,
    manufacture_year int,
    total_km_run decimal(12,2),
    current_status varchar(20) check (current_status IN ('Active', 'Workshop',
'Retired')) NOT NULL
);

```

```

create table operations.train_runs (
    run_id int identity(1,1) primary key,
    train_id int NOT NULL,
    run_date date NOT NULL,
    origin_departure datetime NOT NULL,
    expected_destination_arrival datetime NOT NULL,
    actual_destination_arrival datetime,
    current_location varchar(100),
    delay_minutes int DEFAULT 0,
    run_status varchar(15) check (run_status IN ('OnTime', 'Delayed',
'Cancelled', 'Diverted')) NOT NULL,
    loco_id int
);

```

```

create table operations.real_time_tracking (
    tracking_id int identity(1,1) primary key,
    run_id int NOT NULL,
    recorded_at datetime NOT NULL,
    current_station_id int,
    next_station_id int,
    speed_kmph decimal(5,2),
    latitude decimal(10,7),
    longitude decimal(10,7),
    delay_minutes int default 0
);

```

```

create table operations.coaches (
    coach_id int identity(1,1) primary key,
    train_id int NOT NULL,
    coach_number varchar(10) NOT NULL,
    coach_type varchar(10) check (coach_type IN ('Sleeper', 'AC', 'Pantry',
'Guard')) NOT NULL,
    class_id int,
    total_seats int,

```

```

    coach_position int,
    manufacture_year int,
    last_overhauled date,
    coach_status varchar(10) check (coach_status IN ('Active', 'Workshop',
'Retired')) NOT NULL
);

```

```

create table operations.seats (
    seat_id int identity(1,1) primary key,
    coach_id int NOT NULL,
    seat_number varchar(10) NOT NULL,
    berth_type varchar(10) check (berth_type IN ('LB', 'MB', 'UB', 'SL',
'SU', 'Window')) NOT NULL,
    side_berth bit,
    coupe_number int,
    is_disabled_friendly bit,
    seat_status varchar(10) check (seat_status IN ('Active', 'Damaged',
'Blocked')) NOT NULL
);

```

```

create table operations.platform_allocations (
    allocation_id int identity(1,1) primary key,
    run_id int NOT NULL,
    station_id int NOT NULL,
    platform_id int NOT NULL,
    allocated_at datetime,
    arrival_time datetime,
    departure_time datetime,
    allocation_status varchar(15) check (allocation_status IN ('Allocated',
'Changed', 'Vacated')) NOT NULL
);

```

```

create table staff.departments (
    department_id int identity(1,1) primary key,
    department_name varchar(100) not null,
    department_code varchar(10) not null,
    head_employee_id int,
    zone_id int,
    description text
);

```

```

create table staff.employees (
    employee_id int identity(1,1) primary key,
    employee_code varchar(20) not null unique,
    full_name varchar(150) not null,
    date_of_birth date,

```



```

gender      varchar(10) check (gender in ('m', 'f', 'other')) not null,
mobile      varchar(15),
email       varchar(100),
department_id int,
designation  varchar(100),
joining_date date,
zone_id     int,
employee_status varchar(15) check (employee_status in ('active', 'retired',
'suspended')) not null,
cnic_number  varchar(20)
);

```

```

create table staff.train_crew (
    crew_id      int identity(1,1) primary key,
    run_id       int          not null,
    employee_id  int          not null,
    role         varchar(20) check (role in ('driver', 'guard', 'tte', 'pantry staff'))
not null,
    from_station_id int,
    to_station_id  int,
    duty_start    datetime,
    duty_end      datetime,
    duty_status   varchar(15) check (duty_status in ('assigned', 'completed',
'absent')) not null
);

```

```

create table staff.salaries (
    salary_id     int identity(1,1) primary key,
    employee_id   int          not null,
    month         date         not null,
    basic_salary  decimal(10,2) not null,
    allowances    decimal(8,2) default 0,
    deductions    decimal(8,2) default 0,
    net_salary    decimal(10,2) not null,
    paid_on       date,
    payment_status varchar(10) check (payment_status in ('paid', 'pending',
'hold')) not null
);

```

```

create table staff.leaves (
    leave_id      int identity(1,1) primary key,
    employee_id   int          not null,
    leave_type     varchar(10) check (leave_type in ('cl', 'sl', 'el', 'medical')) not
null,
    from_date     date         not null,
    to_date       date         not null,

```

```

    total_days    int          not null,
    reason        text,
    approved_by   int,
    leave_status  varchar(10) check (leave_status in ('pending', 'approved',
'rejected')) not null
);

```

```

create table maintenance.maintenance_schedules (
    maintenance_id int identity(1,1) primary key,
    asset_type      varchar(20) check (asset_type in
('train','coach','loco','track','station')) not null,
    asset_id        int          not null,
    maintenance_type varchar(20) check (maintenance_type in
('routine','annual','emergency')) not null,
    scheduled_date  date,
    completed_date  date,
    assigned_to     int,
    cost            decimal(10,2),
    maintenance_status varchar(20) check (maintenance_status in
('scheduled','inprogress','completed')) not null,
    remarks         text
);

```

```

create table maintenance.incidents (
    incident_id    int identity(1,1) primary key,
    run_id         int          not null,
    incident_type   varchar(30) check (incident_type in
('accident','derailment','signal failure','fire')) not null,
    occurred_at    datetime,
    location_description text,
    casualties     int          default 0,
    injuries       int          default 0,
    description     text,
    reported_by    int,
    inquiry_status  varchar(20) check (inquiry_status in ('open','under
inquiry','closed')) not null
);

```

```

create table maintenance.track_sections (
    section_id     int identity(1,1) primary key,
    from_station_id int          not null,
    to_station_id  int          not null,
    distance_km    decimal(6,2),
    track_type      varchar(5) check (track_type in ('bg','mg','ng')) not null,
    electrified     bit          default 0,
    max_speed_allowed int,

```

```

    number_of_tracks int,
    last_inspected date,
    section_status varchar(30) check (section_status in ('operational','blocked
for repair')) not null
);

```

```

create table maintenance.complaints (
    complaint_id int identity(1,1) primary key,
    passenger_id int not null,
    booking_id int,
    complaint_type varchar(30) check (complaint_type in
('hygiene','food','staff behavior','safety')) not null,
    complaint_text text,
    filed_at datetime,
    assigned_to int,
    resolved_at datetime,
    resolution_note text,
    complaint_status varchar(20) check (complaint_status in ('open','in
progress','resolved')) not null
);

```

```

create table cargo.freight_bookings (
    freight_id int identity(1,1) primary key,
    consignment_number varchar(20) not null unique,
    sender_name varchar(150) not null,
    receiver_name varchar(150) not null,
    from_station_id int not null references core.stations(station_id),
    to_station_id int not null references core.stations(station_id),
    cargo_type varchar(20) check (cargo_type in
('general','perishable','hazardous')) not null,
    weight_kg decimal(10,2),
    volume_cbm decimal(8,2),
    freight_charge decimal(10,2),
    booked_on datetime,
    delivery_status varchar(20) check (delivery_status in ('booked','in
transit','delivered')) not null
);

```

```

create table cargo.goods_wagons (
    wagon_id int identity(1,1) primary key,
    wagon_number varchar(20) not null,
    wagon_type varchar(15) check (wagon_type in
('flat','box','tank','hopper')) not null,
    capacity_kg decimal(10,2),
    volume_cbm decimal(8,2),
    manufacture_year int,
    current_location int,

```

```
    wagon_status    varchar(15) check (wagon_status in
('available','loaded','workshop')) not null
);
```

```
create table cargo.parcel_services (
    parcel_id        int identity(1,1) primary key,
    train_id         int          not null,
    run_id           int          not null,
    sender_name      varchar(150) not null,
    receiver_contact varchar(15),
    from_station_id  int          not null,
    to_station_id    int          not null,
    weight_kg        decimal(6,2),
    parcel_charge    decimal(8,2),
    delivery_status  varchar(15) check (delivery_status in
('booked','dispatched','delivered')) not null
);
```

```
create table amenities.food_orders (
    order_id         int identity(1,1) primary key,
    ticket_id        int          not null,
    run_id           int          not null,
    station_id       int          not null,
    order_time       datetime,
    total_amount     decimal(8,2),
    food_provider    varchar(100),
    delivery_status  varchar(15) check (delivery_status in
('placed','confirmed','delivered')) not null
);
```

```
create table amenities.food_items (
    item_id          int identity(1,1) primary key,
    item_name        varchar(100) not null,
    category         varchar(15) check (category in ('veg','non-
veg','beverage','snack')) not null,
    price            decimal(7,2),
    available_on_trains text,
    is_available     bit          default 1,
    calories         int
);
```

```
create table amenities.station_facilities (
    facility_id      int identity(1,1) primary key,
    station_id       int          not null,
    facility_type     varchar(20) check (facility_type in
('lounge','wifi','atm','lift','parking','toilet')) not null,
```

```

location_in_station varchar(100),
is_operational      bit          default 1,
operator_name       varchar(100),
charge_applicable   bit          default 0,
charge_amount       decimal(6,2)
);

```

```

create table amenities.passes (
    pass_id      int identity(1,1) primary key,
    passenger_id int          not null,
    pass_type     varchar(15) check (pass_type in ('monthly','quarterly','annual'))
not null,
    from_station_id int      not null,
    to_station_id  int       not null,
    class_id       int       not null,
    valid_from     date,
    valid_to       date,
    pass_amount    decimal(8,2),
    is_active      bit        default 1
);

```

```

create table amenities.lost_and_found (
    lost_id      int identity(1,1) primary key,
    passenger_id int          not null,
    run_id       int          not null,
    coach_number varchar(10),
    item_description text,
    reported_at  datetime,
    found_at_station_id int,
    claim_status varchar(15) check (claim_status in
('lost','found','claimed','disposed')) not null
);

```

ER Diagram: